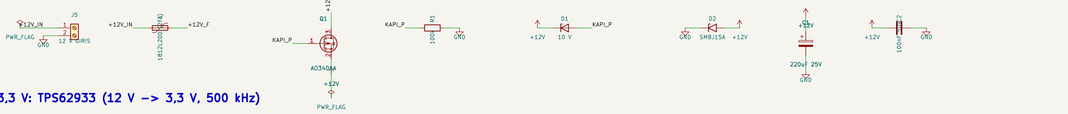
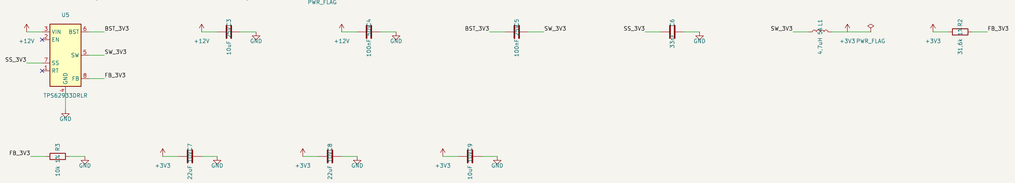
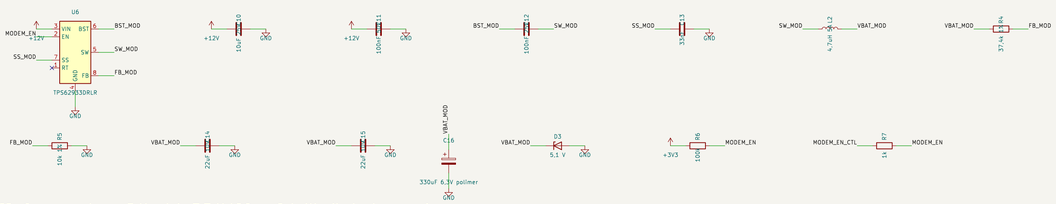
## 12 V GIRISI: klemens, sigorta, ters baglanti ve asiri gerilim korumasi



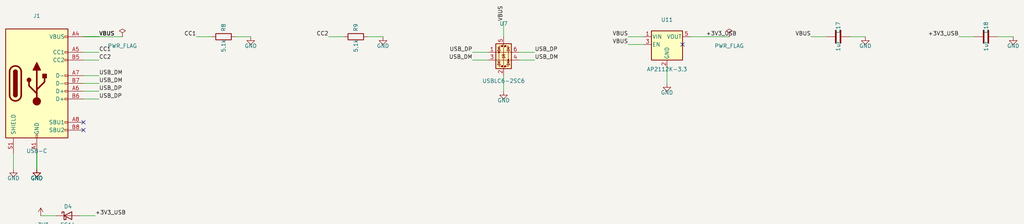

## 3,3 V: TPS62933 (12 V –> 3,3 V, 500 kHz)



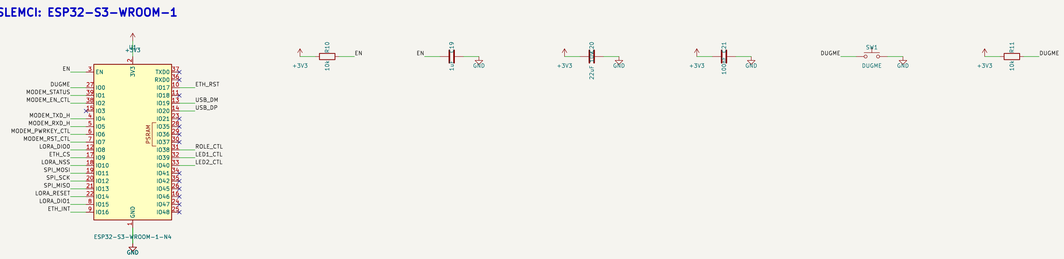

## MODEM BESLEMESI: TPS62933 (12 V –> 3,8 V, >= 2 A tepe) ve toplu kondansator




## USB-C: programlama; 5 V yalniz 3,3 V LDO ve Schottky ile beslemeye girer




## ISLEMCI: ESP32-S3-WROOM-1




## LoRa: RFM95W-915S2




## ETHERNET: W5500 (WIZnet referans devresi, RJ45 + manyetik)

